

1. Euclid's algorithm computes the greatest common divisor of two positive given integers:

```
def euclid_gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a

num1 = int(input("first integer: "))
num2 = int(input("second integer: "))

gcd = euclid_gcd(num1, num2)
print("GDC(", num1, ", ", num2, ")=", gcd)
```

2. PYTHON program to divide the two given integers using subtraction operator:

```
def divide_using_subtraction(dividend, divisor):
    quotient = 0
    while dividend >= divisor:
        dividend -= divisor
        quotient += 1
    return quotient

dividend = int(input(" dividend: "))
divisor = int(input(" divisor: "))

quotient = divide_using_subtraction(dividend, divisor)
print("quotient:", quotient)
```

3. PYTHON program to multiply two given integers without using the multiply operator(*).

```
def multiply_without_operator(num1, num2):
    product = 0
    if num1 < 0 and num2 < 0:
        num1 = -num1
        num2 = -num2
    elif num2 < 0:
        num1, num2 = num2, num1

    while num2 != 0:
        product += num1
        num2 -= 1

    return product
num1 = int(input("first integer: "))
num2 = int(input("second integer: "))

result = multiply_without_operator(num1, num2)
print("Product:", result)
```

4. PYTHON program to accept a positive number and repeatedly add all its digits until the result has only one digit.

```
def sum_digits(num):
    while num >= 10:
        digit_sum = 0
        while num > 0:
            digit_sum += num % 10
            num //= 10
        num = digit_sum
    return num
num = int(input("positive number: "))
result = sum_digits(num)
print("Result:", result)

result = multiply_without_operator(num1, num2)
print("Product:", result)
```

5. PYTHON program that, when you read an integer with 5 digit, display the middle digit.

```
def display_middle_digit(num):
    num_str = str(num)
    middle_index = len(num_str) // 2
    middle_digit = int(num_str[middle_index])

    return middle_digit

num = int(input("enter five digit integer: "))

middle_digit = display_middle_digit(num)
print("middle Digit:", middle_digit)
```

6. PYTHON program that accepts an integer (n) and computes the value of $n + nn + nnn$

```
def compute_value(n):
    nn = int(str(n) + str(n))
    nnn = int(str(n) + str(nn))
    return n + nn + nnn

n = int(input("enter an integer: "))

result = compute_value(n)
print("Result:", result)
```

7. PYTHON program to print numbers between 1 to 100 which are divisible by 3, 5 and by both.

```
for num in range(1, 101):
    if num % 3 == 0 and num % 5 == 0:
        print(num, "is divisible by both 3 and 5")
    elif num % 3 == 0:
        print(num, "is divisible by 3")
    elif num % 5 == 0:
        print(num, "is divisible by 5")
```

8. PYTHON program to compute the sum of the first 100 prime numbers.

```
def is_prime(num):
    if num < 2:
        return False
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            return False
    return True

count = 0
sum_primes = 0
num = 2

while count < 100:
    if is_prime(num):
        sum_primes += num
        count += 1
    num += 1

print("Sum of the first 100 prime numbers:", sum_primes)
```

9. PYTHON program to count the number of days from the beginning of a year when a data is given for the same year.

```
def is_leap_year(year):
    if year % 4 == 0:
        if year % 100 == 0:
            if year % 400 == 0:
                return True
            else:
                return False
        else:
            return True
    else:
        return False
```

```

def count_days(y, m, d):
    days_per_month = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31]
    if is_leap_year(y):
        days_per_month[1] = 29

    total_days = d
    for i in range(m - 1):
        total_days += days_per_month[i]
    return total_days

date = input("enter the date (YYYY-MM-DD): ")
y, m, d = map(int, date.split("-"))

days = count_days(y, m, d)
print("number of days from the beginning of the year:", days)

```

10. PYTHON program to print average of prime numbers between given two integers.

```

def is_prime(num):
    if num < 2:
        return False
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            return False
    return True

def average_of_primes(start, end):
    prime_sum = 0
    count = 0
    for num in range(start, end + 1):
        if is_prime(num):
            prime_sum += num
            count += 1
    if count != 0:
        average = prime_sum / count
        return average
    else:
        return None

start = int(input("Enter the starting integer: "))
end = int(input("Enter the ending integer: "))

average = average_of_primes(start, end)

if average is not None:
    print("average of prime numbers:", average)
else:
    print("no prime numbers found ")

```

